# Simple-opponent results

Shared development benchmark • 10 pilots × 100 matched cases • seed 211307

| Population | Elimination | Survival | Clean win | Mutual loss |
|---|---|---|---|---|
| Original | 59.8% | 57.6% | 28.2% | 31.6% |
| Reduced learning rate | 60.8% | 66.5% | 36.8% | 24.0% |
| Selected-skill probabilities | 72.2% | 68.8% | 50.0% | 22.2% |
| Per-skill parameter heads | 65.4% | 68.3% | 42.5% | 22.9% |

Lower learning rate: the main observed gain was survival, with similar elimination frequency.

Selected-skill probabilities: more eliminations, with a smaller further improvement in survival.

Elimination and survival overlap. Clean win records both occurring in the same flight.

# Experimental development

Independent population training, controlled PPO experiments and tactical adaptation

| | |
|---|---|
| **04 → 05** | One PPO pilot became an arbitrary-size population with separate rewards and checkpoints. |
| **Reward diversity** | Monotonic sweep → relative Halton profiles → seeded, independent uniform ranges. |
| **Evaluation design** | Recorded training/validation and a fixed benchmark; final testing after development. |
| **Optimisation** | Lower learning rate and selected-skill probabilities improved clean wins. |
| **Bounded trials** | More parameter heads underperformed; the terminal-bonus trial stopped incomplete. |
| **Tactical** | The stronger shared-head population was retained and adapted to scripted tactical. |

# Scenario collection and PPO updates

75 scenarios per epoch for each pilot; shared scenario schedules support matched comparisons

## OBSERVE

20 frames × 51 features
2 seconds of flight history

→

## CHOOSE

One of 33 skills
+ its numeric parameters

→

## SIMULATE

Execute about 1 second
Record events and rewards

→

## UPDATE

Estimate advantages
PPO minibatches of 256

## Scenarios and minibatches

A scenario is one duel with seeded starting geometry, speeds and altitudes (up to 120 seconds).
75 scenarios produce many decision transitions. A PPO minibatch contains 256 transitions.
150 epochs × 75 scenarios = 11,250 training flights per pilot; 112,500 across the population.

# Actor–critic network architecture

Shared transformer: width 64 • 4 attention heads • 2 layers • shared parameter predictor

## ACTOR: policy distribution

Predict a distribution over 33 skills
and 16 numeric parameter slots.
Only parameters used by the chosen
skill count in its PPO probability.

## CRITIC: value estimation

Predict future discounted reward.
Compare rewards + future value with
that prediction to estimate advantage.
Positive advantage favours the action.

$$\log \pi(a \mid s) = \log \pi(k \mid s) + \sum_{j \in \text{active}(k)} \log p(z_j \mid s)$$

The selected-skill correction changed the learning objective, without adding network capacity.

# Reward profiles and common evaluation metrics

Each pilot draws its coefficients once using a reproducible random seed (42)

| | |
|---|---|
| Crash / being shot down | Each independently uniform: −450 to −300 |
| Lock / launch with lock | Each independently uniform: 0 to 2 |
| Avoid an incoming missile | Uniform: 0 to 15 |
| Launch without lock | Uniform: −12 to −4 |
| Opponent destroyed / clearance | Fixed: +250 / +0.02 |

## Common outcome scoring

Clean win +100 • survive without elimination 0 • mutual destruction −50 • own loss −100

# Interpretation of simple-opponent results

Changes on the fixed development benchmark • percentage points relative to the matching baseline

## LOWER LEARNING RATE

Eliminations: +1.0 points.
Survival: +8.9 points.
Clean wins: +8.6 points.
Mutual destruction: −7.6 points.

## SELECTED-SKILL PROBABILITIES

Eliminations: +11.4 points.
Survival: +2.3 points.
Clean wins: +13.2 points.
Mutual destruction: −1.8 points.

## Bounded trials

Per-skill heads: eliminations fell 72.2% → 65.4%; survival was similar, 68.8% → 68.3%.
Bonus trial (6 matched pilots): eliminations 72.8% → 72.2%; survival 68.3% → 63.8%.
Both scored fewer clean wins; the shared-head selected-skill population was retained.

# Recorded training and validation

Population mean own return • identical reward profiles across runs • 150 epochs per pilot




Own return describes learning. Assess combat using elimination, survival, clean wins and mutual destruction.

# Tactical adaptation: experimental design

Same reward profiles, skills, physics and PPO settings; new scripted opponent behaviour

## BEFORE

Freeze the 10 selected-skill parents.
Assess all on 300 tactical cases.
Keep the originals unchanged.

## FINE-TUNE

Load each pilot's actor + critic.
Fresh Adam; LR 0.00003.
150 epochs × 75 fresh tactical flights.

## SELECT

50 separate validation cases.
Own reward selects the checkpoint;
initial weights are also eligible.

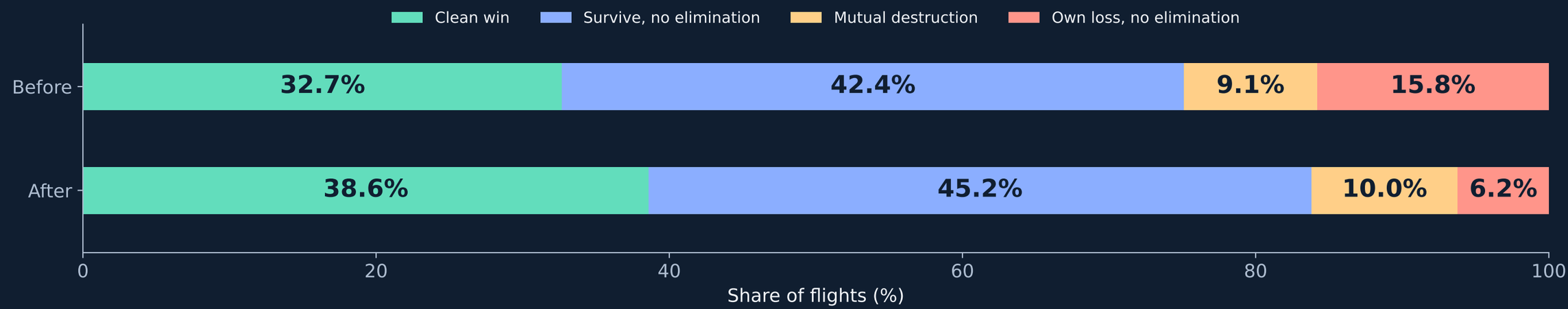
## COMPARE

Repeat the same 300-case assessment.
Use a separate 100-case confirmation
for both parent and adapted pilots.

# Tactical adaptation outcomes

Historical 100-case confirmation • seed 1220000 • the same cases before and after adaptation

| Population | Elimination | Survival | Clean win | Mutual loss |
|---|---|---|---|---|
| Before | 41.8% | 75.1% | 32.7% | 9.1% |
| After | 48.6% | 83.8% | 38.6% | 10.0% |




Clean-win gain: +5.9 pp | Paired scenario bootstrap 95% interval: +1.2 to +11.0 pp

# Per-pilot confirmation results

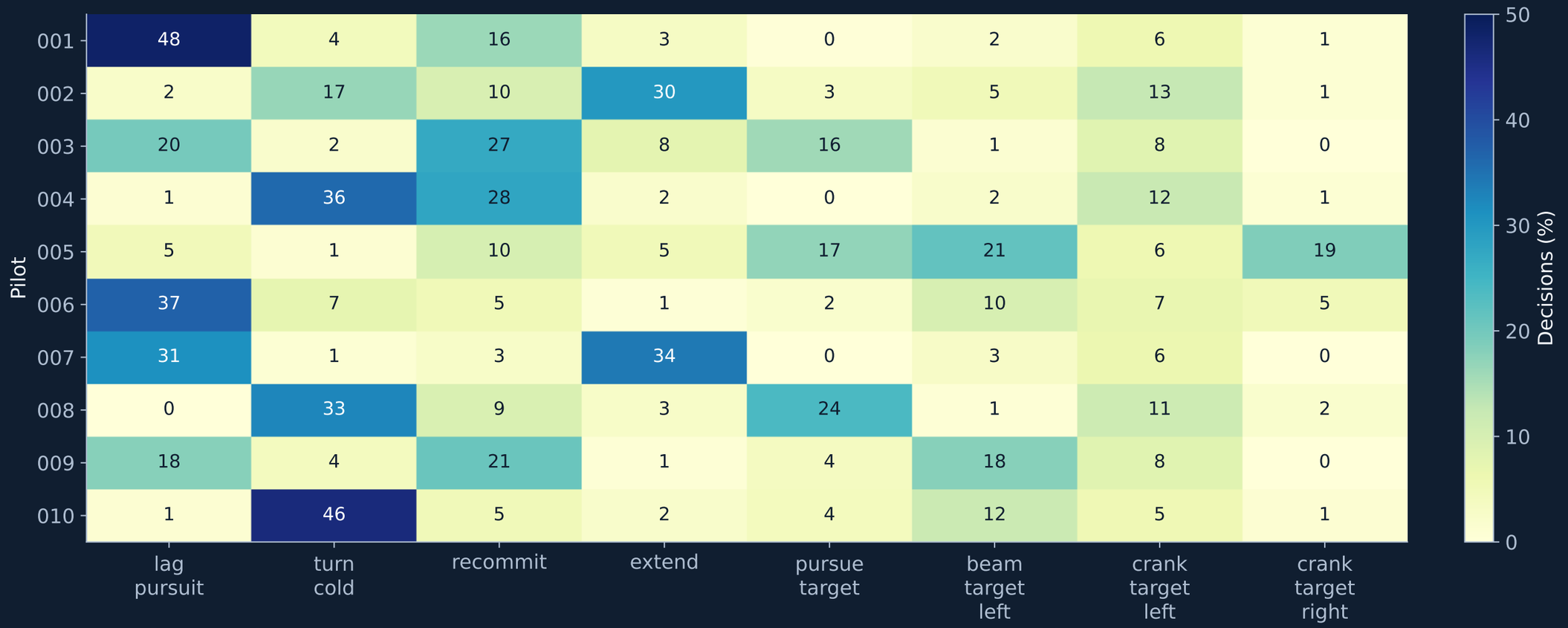
Nine clean-win rates improved and one tied on the same 100 tactical confirmation cases

| Pilot | Elimination | Survival | Clean win | Mutual loss |
|---|---|---|---|---|
| 001 | 42% → 45% | 72% → 81% | 34% → 34% | 8% → 11% |
| 002 | 45% → 49% | 75% → 84% | 32% → 38% | 13% → 11% |
| 003 | 41% → 46% | 74% → 85% | 32% → 37% | 9% → 9% |
| 004 | 42% → 49% | 79% → 84% | 33% → 39% | 9% → 10% |
| 005 | 38% → 52% | 74% → 86% | 31% → 44% | 7% → 8% |
| 006 | 37% → 49% | 65% → 83% | 30% → 39% | 7% → 10% |
| 007 | 43% → 51% | 80% → 85% | 33% → 43% | 10% → 8% |
| 008 | 51% → 50% | 76% → 83% | 37% → 38% | 14% → 12% |
| 009 | 41% → 44% | 76% → 85% | 34% → 35% | 7% → 9% |
| 010 | 38% → 51% | 80% → 82% | 31% → 39% | 7% → 12% |

005 after adaptation: 52% eliminations, 86% survival, 44% clean wins. Small gaps remain uncertain.

# Behavioural variation after adaptation

Observed decision frequencies on confirmation flights • descriptive measures, not aggression rankings




001 favours lag pursuit; 010 often turns cold; 005 mixes beaming, cranking and pursuit.

# Generalisation and evaluation limitations

Matched performance improvements and the scope of the available evidence

## OBSERVED RESULTS

100-case confirmation: +5.9 pp clean wins.
300-case assessment: +6.9 pp clean wins.
Survival improved in both suites.
All 10 training runs completed; parents and frozen source files verified unchanged.

## EVALUATION LIMITATIONS

One training seed per reward profile.
No post-adaptation test against simple.
Mutual destruction did not improve on the confirmation suite.
No universal best pilot or style proven.

No direct seed overlap was found across training, selection and tactical report suites.

Held-out gains do not establish the absence of overfitting to this simulator or opponent.

# Portable reports and supporting data

PDF and self-contained HTML formats; no notebook runtime, Python, GPU or LFS download required

## REPORTS

presentation.pdf — slide deck
presentation.html — offline browser deck
README.md — technical report

## DATA

data/results.json — verified summaries
data/learning_curves.csv — recorded epochs
data/development_summary.csv — pilot results

## Principal findings

Controlled comparisons improved PPO performance in a diverse pilot population. Subsequent tactical adaptation improved clean wins and survival on separate evaluation cases, with continued variation in observed behaviour.